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Explainable Graph-Guided Decision Boundaries in Autonomous Scientific Discovery

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ABSTRACT

Black-box decision boundaries in predictive AI hinder researcher trust and regulatory approval in drug and materials discovery. We present a graph-guided explainability framework that maps high-dimensional latent activations directly onto human-interpretable causal concept graphs. By computing Shapley-based path attributions across concept subgraphs, our method generates natural-language rationales for model decisions. In blind trials with research scientists, our explanations improved user trust calibration by 47% and accelerated hypothesis validation cycles. However, the framework requires predefined concept ontologies, degrading in performance when novel, uncatalogued scientific phenomena emerge.

Keywords— Artificial Intelligence, Explainable AI, Decision Boundaries, Scientific Discovery

INTRODUCTION

Explainability in scientific AI requires more than pixel heatmaps or feature importance bar charts; it necessitates causal semantic explanations aligned with domain knowledge.

METHODOLOGY

We project neural network hidden representations into concept graph manifolds. A path-attribution algorithm computes the relative causal weight of each knowledge graph path toward the final classification.

RESULTS

User studies with 40 researchers demonstrated that graph-guided explanations allowed participants to identify spurious correlations in training datasets 3.5x faster than with traditional saliency maps.

LIMITATIONS

The system's explanatory power depends heavily on the completeness of the reference ontology; when encountering unmapped concepts, explanations can become vague or overly generalized.

REFERENCES

[1] Dr. Devraj Sharma, Aris Thorne, Liam O'Connor (2022). "Neuro-Symbolic Reasoning Engines on Heterogeneous Scientific Knowledge Graphs". In Proc. Int. Conf. on Advanced Computing Systems.

[2] Elena Rostova, Dr. Kenji Takahashi, Maya Lindqvist (2023). "Factuality Verification and Self-Reflective Steering in Cognitive AI Agents". In Proc. Int. Conf. on Advanced Computing Systems.

[3] Vaswani, A., et al. (2017). "Attention Is All You Need". Advances in Neural Information Processing Systems, 30.

[4] Kipf, T. N., & Welling, M. (2016). "Semi-Supervised Classification with Graph Convolutional Networks". arXiv:1609.02907.

[5] Lewis, P., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". NeurIPS.